

Lexical Grammar

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1 Regex

1.1 Substitutions

L	$:=$	$[a - z]$	(1)
U	$:=$	$[A - Z]$	(2)
D	$:=$	$[0 - 9]$	(3)
H	$:=$	$[a - fA - F0 - 9]$	(4)
O	$:=$	$[0 - 7]$	(5)
B	$:=$	$[01]$	(6)

1.2 Identifiers, Constants, and Comments

$_{-}\{0, 2\}L(? : L U D _{-})^{*}$	$\rightarrow$	IDENT	(7)
$U(? : L U D)^{*}$	$\rightarrow$	TYPE_IDENTIFIER	(8)
$[\backslash + -]?D^{+}$	$\rightarrow$	NUMERIC_CONST Integer	(9)
$0[xX]H\{1, 16\}$	$\rightarrow$	NUMERIC_CONST Hex	(10)
$0[oO]O\{1, 32\}$	$\rightarrow$	NUMERIC_CONST Octal	(11)
$0[bB]B\{1, 64\}$	$\rightarrow$	NUMERIC_CONST Binary	(12)
$[\backslash + -]?D + [eE][\backslash + -]?D^{+}$	$\rightarrow$	NUMERIC_CONST Scientific Notation	(13)
$[\backslash + -]?(? : D + (? : [.]D^{*})?[.]D^{+})$	$\rightarrow$	NUMERIC_CONST Float	(14)
$r?"((? : [\w]"\\ \\")^{*})"$	$\rightarrow$	STRING_LITERAL	(15)
$\#. * \backslash n\$$	$\rightarrow$	COMMENT	(16)

1.3 Symbols

,	$\rightarrow$	COMMA	(17)
;	$\rightarrow$	SEMI	(18)
(	$\rightarrow$	LEFT_PAREN	(19)
)	$\rightarrow$	RIGHT_PAREN	(20)
[	$\rightarrow$	LEFT_SQUARE	(21)
]	$\rightarrow$	RIGHT_SQUARE	(22)
{	$\rightarrow$	LEFT_CURLY	(23)
}	$\rightarrow$	RIGHT_CURLY	(24)

1.4 Keywords

and	→	LOGICAL_AND	(25)
bool	→	BOOL	(26)
break	→	BREAK	(27)
class	→	CLASS	(28)
elif	→	ELIF	(29)
else	→	ELSE	(30)
enum	→	ENUM	(31)
exit	→	EXIT	(32)
false	→	FALSE	(33)
float32	→	FLOAT32	(34)
float64	→	FLOAT64	(35)
for	→	FOR	(36)
func	→	FUNCTION	(37)
if	→	IF	(38)
iface	→	INTERFACE	(39)
in	→	IN	(40)
int16	→	INT16	(41)
int32	→	INT32	(42)
int64	→	INT64	(43)
int8	→	INT8	(44)
mut	→	MUTABLE	(45)
or	→	LOGICAL_OR	(46)
private	→	PRIVATE	(47)
public	→	PUBLIC	(48)
string	→	STRING	(49)
true	→	TRUE	(50)
uint16	→	UINT16	(51)
uint32	→	UINT32	(52)
uint64	→	UINT64	(53)
uint8	→	UINT8	(54)
while	→	WHILE	(55)

1.5 Operators

$\wedge\wedge=$	→	EXPONENT_ASSIGN	(56)
$\wedge\wedge$	→	EXPONENT	(57)
$\wedge=$	→	XOR_ASSIGN	(58)
$\wedge$	→	XOR	(59)
$\% =$	→	MOD_ASSIGN	(60)
$\%$	→	MOD	(61)
$* =$	→	MUL_ASSIGN	(62)
$*$	→	MUL	(63)
$// =$	→	INTEGER_DIV_ASSIGN	(64)
$//$	→	INTEGER_DIV	(65)
$/ =$	→	DIV_ASSIGN	(66)
$/$	→	DIV	(67)
$++$	→	INC	(68)
$+ =$	→	ADD_ASSIGN	(69)
$+$	→	ADD	(70)
$--$	→	DEC	(71)
$- =$	→	SUB_ASSIGN	(72)
$-$	→	SUB	(73)
$<<$	→	SHIFT_LEFT	(74)
$<=$	→	LE_OP	(75)
$<$	→	LT_OP	(76)
$>>$	→	SHIFT_RIGHT	(77)
$>=$	→	GE_OP	(78)
$>$	→	GT_OP	(79)
$!=$	→	NE_OP	(80)
$==$	→	EQ_OP	(81)
$=$	→	ASSIGN	(82)
$\& =$	→	AND_ASSIGN	(83)
$\&$	→	AND	(84)
$ =$	→	OR_ASSIGN	(85)
$ $	→	OR	(86)
$?$	→	TERNARY	(87)
$.$	→	DOT	(88)

2 Backus-Naur Form

$\langle \text{ident} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{ident?} \rangle$	(89)
		$ \text{"_"} \langle \text{lower} \rangle \langle \text{ident?} \rangle$	(90)
		$ \text{"_"} \text{"_"} \langle \text{lower} \rangle \langle \text{ident?} \rangle$	(91)
$\langle \text{ident?} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{ident?} \rangle$	(92)
		$ \langle \text{upper} \rangle \langle \text{ident?} \rangle$	(93)
		$ \langle \text{digit} \rangle \langle \text{ident?} \rangle$	(94)
		$ \text{"_"} \langle \text{ident?} \rangle$	(95)
		$ \varepsilon$	(96)
$\langle \text{type_ident} \rangle$	$::=$	$\langle \text{upper} \rangle \langle \text{type_ident?} \rangle$	(97)
$\langle \text{type_ident?} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{type_ident?} \rangle$	(98)
		$ \langle \text{upper} \rangle \langle \text{type_ident?} \rangle$	(99)
		$ \langle \text{digit} \rangle \langle \text{type_ident?} \rangle$	(100)
		$ \varepsilon$	(101)
$\langle \text{float_const} \rangle$	$::=$	$\langle \text{mantissa} \rangle$	(102)
		$ \langle \text{hex_const} \rangle$	(103)
		$ \langle \text{octal_const} \rangle$	(104)
		$ \langle \text{binary_const} \rangle$	(105)
		$ \langle \text{scientific} \rangle$	(106)
$\langle \text{integer_const} \rangle$	$::=$	$\langle \text{decimal_const} \rangle$	(107)
		$ \langle \text{hex_const} \rangle$	(108)
		$ \langle \text{octal_const} \rangle$	(109)
		$ \langle \text{binary_const} \rangle$	(110)
$\langle \text{comment} \rangle$	$::=$	$\text{"\#"} \langle \text{comment_text} \rangle$	(111)
$\langle \text{comment_text} \rangle$	$::=$	$\langle \text{comment_char} \rangle \langle \text{comment_text} \rangle$	(112)
$\langle \text{comment_char} \rangle$	$::=$	$\langle \text{lower} \rangle \langle \text{upper} \rangle \langle \text{digit} \rangle \langle \text{symbol} \rangle \langle \text{non_newline} \rangle$	(113)
$\langle \text{string_literal} \rangle$	$::=$	$\langle \text{r?} \rangle \langle \text{quote} \rangle \langle \text{content} \rangle \langle \text{quote} \rangle$	(114)
$\langle \text{content} \rangle$	$::=$	$\langle \text{char} \rangle \langle \text{content} \rangle \varepsilon$	(115)
$\langle \text{char} \rangle$	$::=$	$\langle \text{escaped_quote} \rangle \langle \text{normal_symbol} \rangle$	(116)
$\langle \text{scientific} \rangle$	$::=$	$\langle \text{sign?} \rangle \langle \text{mantissa} \rangle \langle \text{exponent} \rangle$	(117)
$\langle \text{mantissa} \rangle$	$::=$	$\langle \text{digits} \rangle \text{"."} \langle \text{digits?} \rangle$	(118)
		$ \text{"."} \langle \text{digits} \rangle$	(119)
		$ \langle \text{digits} \rangle$	(120)
$\langle \text{exponent} \rangle$	$::=$	$\text{"e"} \langle \text{decimal_const} \rangle$	(121)
$\langle \text{decimal_const} \rangle$	$::=$	$\langle \text{sign?} \rangle \langle \text{digits} \rangle$	(122)
$\langle \text{hex_const} \rangle$	$::=$	$\text{"0"} \text{"x"} \langle \text{hs} \rangle$	(123)
		$ \text{"0"} \text{"X"} \langle \text{hs} \rangle$	(124)
$\langle \text{octal_const} \rangle$	$::=$	$\text{"0"} \text{"o"} \langle \text{os} \rangle$	(125)
		$ \text{"0"} \text{"O"} \langle \text{os} \rangle$	(126)
$\langle \text{binary_const} \rangle$	$::=$	$\text{"0"} \text{"b"} \langle \text{bs} \rangle$	(127)
		$ \text{"0"} \text{"B"} \langle \text{bs} \rangle$	(128)
			(129)

$\langle \text{digits?} \rangle$	$::= \langle \text{digits} \rangle \mid \varepsilon$	(130)
$\langle \text{digits} \rangle$	$::= \langle \text{digit} \rangle \mid \langle \text{digit} \rangle \langle \text{digits} \rangle$	(131)
$\langle \text{sign?} \rangle$	$::= \langle \text{sign} \rangle \mid \varepsilon$	(132)
$\langle \text{sign} \rangle$	$::= "+" \mid "-"$	(133)
$\langle \text{lower} \rangle$	$::= "a" \mid "b" \mid "c" \mid "d" \mid "e" \mid "f" \mid "g" \mid "h" \mid "i" \mid "j" \mid "k" \mid "l" \mid "m" \mid$ $"n" \mid "o" \mid "p" \mid "q" \mid "r" \mid "s" \mid "t" \mid "u" \mid "v" \mid "w" \mid "x" \mid "y" \mid "z"$	(134)
$\langle \text{upper} \rangle$	$::= "A" \mid "B" \mid "C" \mid "D" \mid "E" \mid "F" \mid "G" \mid "H" \mid "I" \mid "J" \mid "K" \mid "L" \mid "M" \mid$ $"N" \mid "O" \mid "P" \mid "Q" \mid "R" \mid "S" \mid "T" \mid "U" \mid "V" \mid "W" \mid "X" \mid "Y" \mid "Z"$	(135)
$\langle \text{digit} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7" \mid "8" \mid "9"$	(136)
$\langle \text{hex} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7" \mid "8" \mid "9" \mid$ $"a" \mid "b" \mid "c" \mid "d" \mid "e" \mid "f" \mid "A" \mid "B" \mid "C" \mid "D" \mid "E" \mid "F"$	(137)
$\langle \text{octal} \rangle$	$::= "0" \mid "1" \mid "2" \mid "3" \mid "4" \mid "5" \mid "6" \mid "7"$	(138)
$\langle \text{binary} \rangle$	$::= "0" \mid "1"$	(139)
$\langle \text{symbols} \rangle$	$::= \langle \text{normal_symbol} \rangle \mid \langle \text{quote} \rangle \mid \langle \text{backslash} \rangle$	(140)
$\langle \text{normal_symbol} \rangle$	$::= "!" \mid "\#" \mid "$" \mid "\%" \mid "\&" \mid "'" \mid "(" \mid ")" \mid "*" \mid "+" \mid "," \mid "-" \mid "." \mid "/" \mid ":" \mid ";" \mid$ $"<" \mid "=" \mid ">" \mid "?" \mid "@" \mid "[" \mid "]" \mid "[" \mid "]" \mid "_" \mid "`" \mid "{" \mid "}" \mid " " \mid "$	(141)
$\langle \text{whitespace} \rangle$	$::= \langle \text{non_newline} \rangle \mid \langle \text{newline} \rangle$	(142)
$\langle \text{non_newline} \rangle$	$::= " " \mid "\backslash t" \mid "\backslash r"$	(143)
$\langle \text{newline} \rangle$	$::= "\backslash n"$	(144)
$\langle \text{r?} \rangle$	$::= "r" \mid \varepsilon$	(145)
$\langle \text{escaped_quote} \rangle$	$::= \langle \text{backslash} \rangle \langle \text{quote} \rangle$	(146)
$\langle \text{backslash} \rangle$	$::= "\backslash"$	(147)
$\langle \text{quote} \rangle$	$::= "\""$	(148)
		(149)

$\langle hs \rangle$	$::=$	$\langle hex \rangle$	(150)
		$ \langle hex \rangle \langle hs1 \rangle$	(151)
$\langle hs1 \rangle$	$::=$	$\langle hex \rangle$	(152)
		$ \langle hex \rangle \langle hs2 \rangle$	(153)
$\langle hs2 \rangle$	$::=$	$\langle hex \rangle$	(154)
		$ \langle hex \rangle \langle hs3 \rangle$	(155)
$\langle hs3 \rangle$	$::=$	$\langle hex \rangle$	(156)
		$ \langle hex \rangle \langle hs4 \rangle$	(157)
$\langle hs4 \rangle$	$::=$	$\langle hex \rangle$	(158)
		$ \langle hex \rangle \langle hs5 \rangle$	(159)
$\langle hs5 \rangle$	$::=$	$\langle hex \rangle$	(160)
		$ \langle hex \rangle \langle hs6 \rangle$	(161)
$\langle hs6 \rangle$	$::=$	$\langle hex \rangle$	(162)
		$ \langle hex \rangle \langle hs7 \rangle$	(163)
$\langle hs7 \rangle$	$::=$	$\langle hex \rangle$	(164)
		$ \langle hex \rangle \langle hs8 \rangle$	(165)
$\langle hs8 \rangle$	$::=$	$\langle hex \rangle$	(166)
		$ \langle hex \rangle \langle hs9 \rangle$	(167)
$\langle hs9 \rangle$	$::=$	$\langle hex \rangle$	(168)
		$ \langle hex \rangle \langle hs10 \rangle$	(169)
$\langle hs10 \rangle$	$::=$	$\langle hex \rangle$	(170)
		$ \langle hex \rangle \langle hs11 \rangle$	(171)
$\langle hs11 \rangle$	$::=$	$\langle hex \rangle$	(172)
		$ \langle hex \rangle \langle hs12 \rangle$	(173)
$\langle hs12 \rangle$	$::=$	$\langle hex \rangle$	(174)
		$ \langle hex \rangle \langle hs13 \rangle$	(175)
$\langle hs13 \rangle$	$::=$	$\langle hex \rangle$	(176)
		$ \langle hex \rangle \langle hs14 \rangle$	(177)
$\langle hs14 \rangle$	$::=$	$\langle hex \rangle$	(178)
		$ \langle hex \rangle \langle hs15 \rangle$	(179)
$\langle hs15 \rangle$	$::=$	$\langle hex \rangle$	(180)

$\langle os \rangle$	$::=$	$\langle octal \rangle$	(181)
		$ \langle octal \rangle \langle os1 \rangle$	(182)
$\langle os1 \rangle$	$::=$	$\langle octal \rangle$	(183)
		$ \langle octal \rangle \langle os2 \rangle$	(184)
$\langle os2 \rangle$	$::=$	$\langle octal \rangle$	(185)
		$ \langle octal \rangle \langle os3 \rangle$	(186)
$\langle os3 \rangle$	$::=$	$\langle octal \rangle$	(187)
		$ \langle octal \rangle \langle os4 \rangle$	(188)
$\langle os4 \rangle$	$::=$	$\langle octal \rangle$	(189)
		$ \langle octal \rangle \langle os5 \rangle$	(190)
$\langle os5 \rangle$	$::=$	$\langle octal \rangle$	(191)
		$ \langle octal \rangle \langle os6 \rangle$	(192)
$\langle os6 \rangle$	$::=$	$\langle octal \rangle$	(193)
		$ \langle octal \rangle \langle os7 \rangle$	(194)
$\langle os7 \rangle$	$::=$	$\langle octal \rangle$	(195)
		$ \langle octal \rangle \langle os8 \rangle$	(196)
$\langle os8 \rangle$	$::=$	$\langle octal \rangle$	(197)
		$ \langle octal \rangle \langle os9 \rangle$	(198)
$\langle os9 \rangle$	$::=$	$\langle octal \rangle$	(199)
		$ \langle octal \rangle \langle os10 \rangle$	(200)
$\langle os10 \rangle$	$::=$	$\langle octal \rangle$	(201)
		$ \langle octal \rangle \langle os11 \rangle$	(202)
$\langle os11 \rangle$	$::=$	$\langle octal \rangle$	(203)
		$ \langle octal \rangle \langle os12 \rangle$	(204)
$\langle os12 \rangle$	$::=$	$\langle octal \rangle$	(205)
		$ \langle octal \rangle \langle os13 \rangle$	(206)
$\langle os13 \rangle$	$::=$	$\langle octal \rangle$	(207)
		$ \langle octal \rangle \langle os14 \rangle$	(208)
$\langle os14 \rangle$	$::=$	$\langle octal \rangle$	(209)
		$ \langle octal \rangle \langle os15 \rangle$	(210)
$\langle os15 \rangle$	$::=$	$\langle octal \rangle$	(211)
		$ \langle octal \rangle \langle os16 \rangle$	(212)
$\langle os16 \rangle$	$::=$	$\langle octal \rangle$	(213)
		$ \langle octal \rangle \langle os17 \rangle$	(214)
$\langle os17 \rangle$	$::=$	$\langle octal \rangle$	(215)
		$ \langle octal \rangle \langle os18 \rangle$	(216)
$\langle os18 \rangle$	$::=$	$\langle octal \rangle$	(217)
		$ \langle octal \rangle \langle os19 \rangle$	(218)
$\langle os19 \rangle$	$::=$	$\langle octal \rangle$	(219)
		$ \langle octal \rangle \langle os20 \rangle$	(220)

$\langle os20 \rangle$	$::=$	$\langle octal \rangle$	(221)
		$ \langle octal \rangle \langle os21 \rangle$	(222)
$\langle os21 \rangle$	$::=$	$\langle octal \rangle$	(223)
		$ \langle octal \rangle \langle os22 \rangle$	(224)
$\langle os22 \rangle$	$::=$	$\langle octal \rangle$	(225)
		$ \langle octal \rangle \langle os23 \rangle$	(226)
$\langle os23 \rangle$	$::=$	$\langle octal \rangle$	(227)
		$ \langle octal \rangle \langle os24 \rangle$	(228)
$\langle os24 \rangle$	$::=$	$\langle octal \rangle$	(229)
		$ \langle octal \rangle \langle os25 \rangle$	(230)
$\langle os25 \rangle$	$::=$	$\langle octal \rangle$	(231)
		$ \langle octal \rangle \langle os26 \rangle$	(232)
$\langle os26 \rangle$	$::=$	$\langle octal \rangle$	(233)
		$ \langle octal \rangle \langle os27 \rangle$	(234)
$\langle os27 \rangle$	$::=$	$\langle octal \rangle$	(235)
		$ \langle octal \rangle \langle os28 \rangle$	(236)
$\langle os28 \rangle$	$::=$	$\langle octal \rangle$	(237)
		$ \langle octal \rangle \langle os29 \rangle$	(238)
$\langle os29 \rangle$	$::=$	$\langle octal \rangle$	(239)
		$ \langle octal \rangle \langle os30 \rangle$	(240)
$\langle os30 \rangle$	$::=$	$\langle octal \rangle$	(241)
		$ \langle octal \rangle \langle os31 \rangle$	(242)
$\langle os31 \rangle$	$::=$	$\langle octal \rangle$	(243)
$\langle bs \rangle$	$::=$	$\langle binary \rangle$	(244)
		$ \langle binary \rangle \langle bs1 \rangle$	(245)
$\langle bs1 \rangle$	$::=$	$\langle binary \rangle$	(246)
		$ \langle binary \rangle \langle bs2 \rangle$	(247)
$\langle bs2 \rangle$	$::=$	$\langle binary \rangle$	(248)
		$ \langle binary \rangle \langle bs3 \rangle$	(249)
$\langle bs3 \rangle$	$::=$	$\langle binary \rangle$	(250)
		$ \langle binary \rangle \langle bs4 \rangle$	(251)
$\langle bs4 \rangle$	$::=$	$\langle binary \rangle$	(252)
		$ \langle binary \rangle \langle bs5 \rangle$	(253)
$\langle bs5 \rangle$	$::=$	$\langle binary \rangle$	(254)
		$ \langle binary \rangle \langle bs6 \rangle$	(255)
$\langle bs6 \rangle$	$::=$	$\langle binary \rangle$	(256)
		$ \langle binary \rangle \langle bs7 \rangle$	(257)

$\langle bs7 \rangle$	::=	$\langle binary \rangle$	(258)
		$ \langle binary \rangle \langle bs8 \rangle$	(259)
$\langle bs8 \rangle$	::=	$\langle binary \rangle$	(260)
		$ \langle binary \rangle \langle bs9 \rangle$	(261)
$\langle bs9 \rangle$	::=	$\langle binary \rangle$	(262)
		$ \langle binary \rangle \langle bs10 \rangle$	(263)
$\langle bs10 \rangle$	::=	$\langle binary \rangle$	(264)
		$ \langle binary \rangle \langle bs11 \rangle$	(265)
$\langle bs11 \rangle$	::=	$\langle binary \rangle$	(266)
		$ \langle binary \rangle \langle bs12 \rangle$	(267)
$\langle bs12 \rangle$	::=	$\langle binary \rangle$	(268)
		$ \langle binary \rangle \langle bs13 \rangle$	(269)
$\langle bs13 \rangle$	::=	$\langle binary \rangle$	(270)
		$ \langle binary \rangle \langle bs14 \rangle$	(271)
$\langle bs14 \rangle$	::=	$\langle binary \rangle$	(272)
		$ \langle binary \rangle \langle bs15 \rangle$	(273)
$\langle bs15 \rangle$	::=	$\langle binary \rangle$	(274)
		$ \langle binary \rangle \langle bs16 \rangle$	(275)
$\langle bs16 \rangle$	::=	$\langle binary \rangle$	(276)
		$ \langle binary \rangle \langle bs17 \rangle$	(277)
$\langle bs17 \rangle$	::=	$\langle binary \rangle$	(278)
		$ \langle binary \rangle \langle bs18 \rangle$	(279)
$\langle bs18 \rangle$	::=	$\langle binary \rangle$	(280)
		$ \langle binary \rangle \langle bs19 \rangle$	(281)
$\langle bs19 \rangle$	::=	$\langle binary \rangle$	(282)
		$ \langle binary \rangle \langle bs20 \rangle$	(283)
$\langle bs20 \rangle$	::=	$\langle binary \rangle$	(284)
		$ \langle binary \rangle \langle bs21 \rangle$	(285)
$\langle bs21 \rangle$	::=	$\langle binary \rangle$	(286)
		$ \langle binary \rangle \langle bs22 \rangle$	(287)
$\langle bs22 \rangle$	::=	$\langle binary \rangle$	(288)
		$ \langle binary \rangle \langle bs23 \rangle$	(289)
$\langle bs23 \rangle$	::=	$\langle binary \rangle$	(290)
		$ \langle binary \rangle \langle bs24 \rangle$	(291)
$\langle bs24 \rangle$	::=	$\langle binary \rangle$	(292)
		$ \langle binary \rangle \langle bs25 \rangle$	(293)
$\langle bs25 \rangle$	::=	$\langle binary \rangle$	(294)
		$ \langle binary \rangle \langle bs26 \rangle$	(295)
$\langle bs26 \rangle$	::=	$\langle binary \rangle$	(296)
		$ \langle binary \rangle \langle bs27 \rangle$	(297)

$\langle bs27 \rangle$	$::=$	$\langle binary \rangle$	(298)
		$ \langle binary \rangle \langle bs28 \rangle$	(299)
$\langle bs28 \rangle$	$::=$	$\langle binary \rangle$	(300)
		$ \langle binary \rangle \langle bs29 \rangle$	(301)
$\langle bs29 \rangle$	$::=$	$\langle binary \rangle$	(302)
		$ \langle binary \rangle \langle bs30 \rangle$	(303)
$\langle bs30 \rangle$	$::=$	$\langle binary \rangle$	(304)
		$ \langle binary \rangle \langle bs31 \rangle$	(305)
$\langle bs31 \rangle$	$::=$	$\langle binary \rangle$	(306)
		$ \langle binary \rangle \langle bs32 \rangle$	(307)
$\langle bs32 \rangle$	$::=$	$\langle binary \rangle$	(308)
		$ \langle binary \rangle \langle bs33 \rangle$	(309)
$\langle bs33 \rangle$	$::=$	$\langle binary \rangle$	(310)
		$ \langle binary \rangle \langle bs34 \rangle$	(311)
$\langle bs34 \rangle$	$::=$	$\langle binary \rangle$	(312)
		$ \langle binary \rangle \langle bs35 \rangle$	(313)
$\langle bs35 \rangle$	$::=$	$\langle binary \rangle$	(314)
		$ \langle binary \rangle \langle bs36 \rangle$	(315)
$\langle bs36 \rangle$	$::=$	$\langle binary \rangle$	(316)
		$ \langle binary \rangle \langle bs37 \rangle$	(317)
$\langle bs37 \rangle$	$::=$	$\langle binary \rangle$	(318)
		$ \langle binary \rangle \langle bs38 \rangle$	(319)
$\langle bs38 \rangle$	$::=$	$\langle binary \rangle$	(320)
		$ \langle binary \rangle \langle bs39 \rangle$	(321)
$\langle bs39 \rangle$	$::=$	$\langle binary \rangle$	(322)
		$ \langle binary \rangle \langle bs40 \rangle$	(323)
$\langle bs40 \rangle$	$::=$	$\langle binary \rangle$	(324)
		$ \langle binary \rangle \langle bs41 \rangle$	(325)
$\langle bs41 \rangle$	$::=$	$\langle binary \rangle$	(326)
		$ \langle binary \rangle \langle bs42 \rangle$	(327)
$\langle bs42 \rangle$	$::=$	$\langle binary \rangle$	(328)
		$ \langle binary \rangle \langle bs43 \rangle$	(329)
$\langle bs43 \rangle$	$::=$	$\langle binary \rangle$	(330)
		$ \langle binary \rangle \langle bs44 \rangle$	(331)
$\langle bs44 \rangle$	$::=$	$\langle binary \rangle$	(332)
		$ \langle binary \rangle \langle bs45 \rangle$	(333)
$\langle bs45 \rangle$	$::=$	$\langle binary \rangle$	(334)
		$ \langle binary \rangle \langle bs46 \rangle$	(335)
$\langle bs46 \rangle$	$::=$	$\langle binary \rangle$	(336)
		$ \langle binary \rangle \langle bs47 \rangle$	(337)

$\langle bs47 \rangle$	$::=$	$\langle binary \rangle$	(338)
		$ \langle binary \rangle \langle bs48 \rangle$	(339)
$\langle bs48 \rangle$	$::=$	$\langle binary \rangle$	(340)
		$ \langle binary \rangle \langle bs49 \rangle$	(341)
$\langle bs49 \rangle$	$::=$	$\langle binary \rangle$	(342)
		$ \langle binary \rangle \langle bs50 \rangle$	(343)
$\langle bs50 \rangle$	$::=$	$\langle binary \rangle$	(344)
		$ \langle binary \rangle \langle bs51 \rangle$	(345)
$\langle bs51 \rangle$	$::=$	$\langle binary \rangle$	(346)
		$ \langle binary \rangle \langle bs52 \rangle$	(347)
$\langle bs52 \rangle$	$::=$	$\langle binary \rangle$	(348)
		$ \langle binary \rangle \langle bs53 \rangle$	(349)
$\langle bs53 \rangle$	$::=$	$\langle binary \rangle$	(350)
		$ \langle binary \rangle \langle bs54 \rangle$	(351)
$\langle bs54 \rangle$	$::=$	$\langle binary \rangle$	(352)
		$ \langle binary \rangle \langle bs55 \rangle$	(353)
$\langle bs55 \rangle$	$::=$	$\langle binary \rangle$	(354)
		$ \langle binary \rangle \langle bs56 \rangle$	(355)
$\langle bs56 \rangle$	$::=$	$\langle binary \rangle$	(356)
		$ \langle binary \rangle \langle bs57 \rangle$	(357)
$\langle bs57 \rangle$	$::=$	$\langle binary \rangle$	(358)
		$ \langle binary \rangle \langle bs58 \rangle$	(359)
$\langle bs58 \rangle$	$::=$	$\langle binary \rangle$	(360)
		$ \langle binary \rangle \langle bs59 \rangle$	(361)
$\langle bs59 \rangle$	$::=$	$\langle binary \rangle$	(362)
		$ \langle binary \rangle \langle bs60 \rangle$	(363)
$\langle bs60 \rangle$	$::=$	$\langle binary \rangle$	(364)
		$ \langle binary \rangle \langle bs61 \rangle$	(365)
$\langle bs61 \rangle$	$::=$	$\langle binary \rangle$	(366)
		$ \langle binary \rangle \langle bs62 \rangle$	(367)
$\langle bs62 \rangle$	$::=$	$\langle binary \rangle$	(368)
		$ \langle binary \rangle \langle bs63 \rangle$	(369)
$\langle bs63 \rangle$	$::=$	$\langle binary \rangle$	(370)
			(371)